

Oral Exam

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Projects

I will briefly discuss the three following projects:

- ▶ Invariant measures for the Stochastic Heat Equation with rough initial conditions,
- ▶ Exact moment asymptotics and Solvability for the Interpolated Stochastic Heat and Wave Equation,
- ▶ Global solutions for the Interpolated Stochastic Heat and Wave Equation with a super linear diffusion term and no drift.

Invariant Measures for the Stochastic Heat Equation

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2} \Delta_x u(t, x) = b(x, u(t, x)) \dot{W}(t, x) \\ u(0, \cdot) = \mu(\cdot) \end{cases}$$

We assume that:

- ▶ $|b(x, u) - b(x, v)| < L_b |u - v|$ and $|b(x, 0)| \leq L_0$ for $u, v \in \mathbb{R}$ and $x \in \mathbb{R}^d$,
- ▶ W is a centered Gaussian noise that is homogeneously colored in space,

$$J(\phi, \psi) := \mathbb{E} [W(\phi) W(\psi)] = \int_0^\infty ds \int_{\mathbb{R}^d} \Gamma(dx) (\phi(s, \cdot) * \tilde{\psi}(s, \cdot))(x),$$

where $\tilde{\psi}(s, x) = \psi(s, -x)$ and Γ is a non-negative and nonnegative definite measure referred to as the *spatial correlation measure*.

Mild Solution

We interpret the solution of the Stochastic Heat Equation as the following stochastic integral equation:

$$\begin{aligned}u(t, x) &= \int_{\mathbb{R}^d} G(t, x - y) \mu(dy) + \int_0^t \int_{\mathbb{R}^d} G(t - s, x - y) b(u(s, y)) W(ds, dy) \\&= J(t, x) + I(t, x),\end{aligned}$$

where $G(t, x)$ is the heat kernel

$$G(t, x) = (2\pi t)^{-d/2} \exp\left(-\frac{|x|^2}{2t}\right)$$

A processes, $\{u(t, x) : t \geq 0, x \in \mathbb{R}^d\}$, that satisfies this relation is referred to as the *mild solution*.

Existence and Uniqueness

Existence and uniqueness of a mild solution of the Stochastic Heat Equation has been proven under the following two conditions:

- ▶ the initial measure, μ , satisfies

$$\int_{\mathbb{R}^d} \exp(-a|x|^2) \mu(dx) < \infty \quad \text{for any } a > 0,$$

- ▶ the Fourier transform of the spatial correlation measure, $\hat{f} := \mathcal{F}\Gamma$ satisfies *Dalang's condition*:

$$\Upsilon(\beta) := \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{\beta + |\xi|^2} < \infty.$$

We refer to $\hat{f}$ as the spectral measure of the noise $\dot{W}$.

Admissible Weight

Since we work in the whole space $\mathbb{R}^d$, we consider weights, ρ such that

- ▶ $\rho \in L^1(\mathbb{R}^d)$ is positive, continuous and bounded,
- ▶ for any $T > 0$, there exists a $C(T) > 0$ such that

$$\sup_{x \in \mathbb{R}^d, t \in [0, T]} \frac{(G(t, \cdot) * \rho)(x)}{\rho(x)} \leq C(T).$$

We then work in the weighted Hilbert Space, $L^2_\rho(\mathbb{R}^d)$, with norm defined as

$$\|\phi(\cdot)\|_{L^2_\rho(\mathbb{R}^d)}^2 = \int_{\mathbb{R}^d} \phi(x)^2 \rho(x) dx.$$

Intermittency

We define

$$\underline{m}_2 := \liminf_{t>0} \frac{1}{t} \log \left(\mathbb{E}(|u(t, x)|^2) \right), \text{ and } \overline{m}_2 := \limsup_{t>0} \frac{1}{t} \log \left(\mathbb{E}(|u(t, x)|^2) \right).$$

The solution is said to be *fully intermittent* when

$$\inf_{x \in \mathbb{R}^d} \underline{m}_2 > 0.$$

We say a *phase transition* happens when there exists $\underline{\lambda} \leq \overline{\lambda}$ such that

$$\begin{cases} \sup_{x \in \mathbb{R}^d} \overline{m}_2 = 0 & L_b < \underline{\lambda} \\ \inf_{x \in \mathbb{R}^d} \underline{m}_2 = 0 & \overline{\lambda} < l_b \end{cases}$$

The Stochastic Heat and Wave Equations

$$\begin{cases} (\partial_t^b - \Delta) u(t, x) = u(t, x) \dot{W}(x) & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & b = 1 \quad (\text{SHE}) \\ u(0, \cdot) = 1, \quad \partial_t u(0, \cdot) = 0 & b = 2 \quad (\text{SWE}) \end{cases}$$

- ▶ $W = \{W(\phi) : \phi \in \mathcal{D}(\mathbb{R}^d)\}$ is a centered and time-independent Gaussian noise
- ▶ The choice of initial condition is such that the solution to the homogeneous equation is constant one.
- ▶ The solution is understood in the *Skorohod* sense.

For the SHE, i.e., $b = 1$ ($a = 2, \nu = 2$)
[X. Chen '17]

$$\lim_{t \rightarrow \infty} t^{-\frac{4-\alpha}{2-\alpha}} \log \mathbb{E}|u(t, x)|^p = p(p-1)^{\frac{2}{2-\alpha}} (2-\alpha) \left(\frac{2\mathcal{M}}{4-\alpha} \right)^{\frac{4-\alpha}{2-\alpha}}$$

For the SWE, i.e., $b = 2$ ($a = 2, \nu = 2$)
[Balan, L. Chen, and X. Chen '21]

$$\lim_{t \rightarrow \infty} t^{-\frac{4-\alpha}{3-\alpha}} \log \mathbb{E}|u(t, x)|^p = p(p-1)^{\frac{1}{3-\alpha}} \left(\frac{1}{2} \right)^{\frac{\alpha}{2(3-\alpha)}} \frac{3-\alpha}{2} \left(\frac{2\mathcal{M}^{1/2}}{4-\alpha} \right)^{\frac{4-\alpha}{3-\alpha}}$$

For the SHE, i.e., $b = 1$ ($a = 2, \nu = 2$)
[X. Chen '17]

$$\lim_{t \rightarrow \infty} t^{-\frac{4-\alpha}{2-\alpha}} \log \mathbb{E}|u(t, x)|^p = p(p-1)^{\frac{2}{2-\alpha}} (2-\alpha) \left(\frac{2\mathcal{M}}{4-\alpha} \right)^{\frac{4-\alpha}{2-\alpha}}$$

$$b \in (0, 2) \text{ and } a \in (0, 2] ??$$

For the SWE, i.e., $b = 2$ ($a = 2, \nu = 2$)
[Balan, L. Chen, and X. Chen '21]

$$\lim_{t \rightarrow \infty} t^{-\frac{4-\alpha}{3-\alpha}} \log \mathbb{E}|u(t, x)|^p = p(p-1)^{\frac{1}{3-\alpha}} \left(\frac{1}{2} \right)^{\frac{\alpha}{2(3-\alpha)}} \frac{3-\alpha}{2} \left(\frac{2\mathcal{M}^{1/2}}{4-\alpha} \right)^{\frac{4-\alpha}{3-\alpha}}$$

The Interpolated Stochastic Heat and Wave Equation

$$\begin{cases} (\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r \left[\sqrt{\theta} u(t, x) \dot{W}(x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & b \in (0, 1] \\ u(0, \cdot) = 1, \quad \partial_t u(0, \cdot) = 0 & b \in (1, 2) \end{cases} \quad (\text{ISHWE})$$

- ▶ ∂_t^b is the *Caputo* fractional derivative
- ▶ $(-\Delta)^{a/2}$ is the fractional Laplacian of order $a \in (0, 2]$
- ▶ I_t^r is the Riemann-Liouville fractional integral of order $r \geq 0$

The Noise, $\dot{W}$.

The time independent noise informally satisfies

$$\mathbb{E}(\dot{W}(x)\dot{W}(y)) = \gamma(x - y).$$

The spatial correlation and spectral density, γ and φ , for the noise $\dot{W}$ can be assumed to be any of the following:

$$\gamma(x) = |x|^{-\alpha}, \quad \varphi(\xi) = C_1 |\xi|^{d-\alpha} \quad \alpha \in (0, d)$$

$$\gamma(x) = \prod_{i=1}^d |x_i|^{-\alpha_i}, \quad \varphi(\xi) = C_2 \prod_{i=1}^d |\xi_i|^{1-\alpha_i} \quad \alpha_i \in (0, 1), \quad \alpha = \sum_i \alpha_i$$

Mild Solution

Definition

For $T > 0$, a random field $u = \{u(t, x) : t \in (0, T), x \in \mathbb{R}^d\}$ is called a *mild solution* if $G(t - s, x - \cdot)u(s, \cdot)1_{\{s < t\}}$ is Skorohod integrable and the following holds almost surely:

$$u(t, x) = 1 + \sqrt{\theta} \int_0^t \left(\int_{\mathbb{R}^d} G(t - s, x - y) u(s, y) W(\delta y) \right) ds$$

where G is defined through the *Fox-H function*.

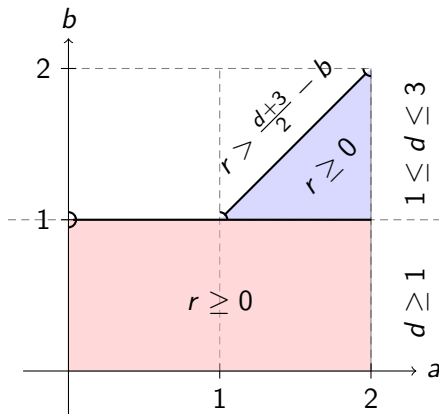
An important characteristic of G is that

$$\mathcal{FL}G(s, \xi) = \frac{s^{-r}}{s^b + \frac{\nu}{2} |\xi|^a}$$

Nonnegativity assumption on G

Under any of the following cases, G is nonnegative [Chen, Hu, Nualart '19]:

- ▶ $d \geq 1$, $b \in (0, 1]$, $a \in (0, 2]$, $r \geq 0$;
- ▶ $1 \leq d \leq 3$, $1 < b < a \leq 2$, $r > 0$;
- ▶ $1 \leq d \leq 3$, $1 < b = a < 2$, $r > \frac{d+3}{2} - b$.



Global Solution

$$(\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r \left[\sqrt{\theta} u(t, x) \dot{W}(x) \right]$$

Definition

$u(t, x)$ is a *global solution* to the (ISHWE) if $\|u(t, x)\|_p < \infty$ for any $t > 0$ and $x \in \mathbb{R}^d$.

Theorem (Chen-E. '21+)

A global solution exists provided G is nonnegative and

$$0 < \alpha < \min \left(\frac{a}{b} [2(b+r) - 1], 2a, d \right)$$

Local Solution

Definition

$u(t, x)$ is a *local solution* to the (ISHWE) if there exists $0 < T_a \leq T_b < \infty$ such that $\|u(t, x)\|_2 < \infty$ when $0 < t < T_a$ and $\|u(t, x)\|_2$ D.N.E. (does not exist) for $t > T_b$.

Theorem (Chen-E. '21+)

A local solution exists provided G is nonnegative and if

$$r \in [0, 1/2] \quad \text{and} \quad 0 < \alpha = \frac{a}{b}[2(b+r) - 1] \leq d.$$

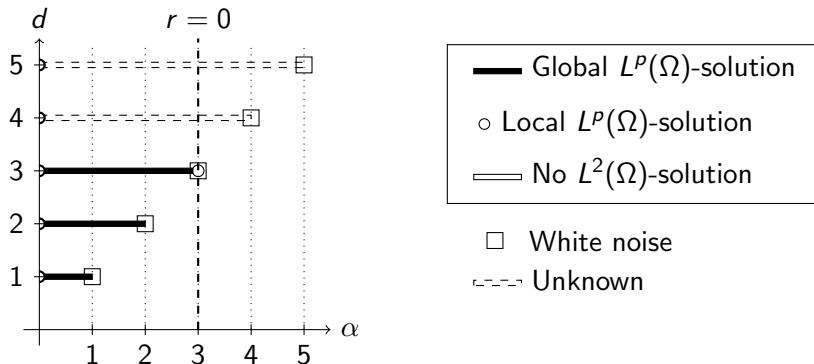
In this case, a unique $L^p(\Omega)$ solution exists for $t \in (0, T_p)$ where

$$T_p := \frac{\nu^{\alpha/a}}{2\theta(p-1)\mathcal{M}_a^{(2a-\alpha)/a}}$$

and the solution does not exist for $t > T_2$.

Example: Solvability for the SWE ($a = b = 2$)

A local solution only exists when $\alpha = 3 + 2r \leq d \leq 3$.

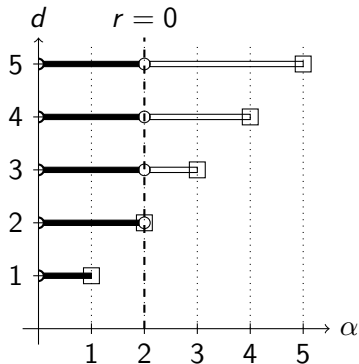


By replacing $\nu = 2$ in the following, we recover (1.12) [Balan, L. Chen & X. Chen '21]

$$T_p = \frac{\nu^{3/2}}{2\theta(p-1)\sqrt{\mathcal{M}_{2,3}(\delta_0)}}, \quad p \geq 2$$

Example: Solvability for the SHE ($a = 2$ and $b = 1$)

By setting $a = 2$, $b = 1$ and $r = 0$, we obtain the following condition for existence of a local solution: $\alpha = 2 \leq d$.



- ▶ When $\alpha = d = 2$, the critical time becomes
$$T_p = \frac{\nu}{2\theta(p-1)\mathcal{M}_{2,2}(\delta_0)}.$$
- ▶ Theorem 4.1 [Y. Hu, '01] proves that an $L^2(\Omega)$ solution exists for $t < 2$ but not for $t > 2\pi$.
- ▶ T_2 being precise implies
$$2 \leq T_2 = \frac{1}{2\mathcal{M}_{2,2}(\delta_0)} \leq 2\pi$$

Wiener Chaos Expansion

Through a standard procedure, we may define

$$f_n(x_1, \dots, x_n; x, t) = \int_0^t \int_0^{t_n} \cdots \int_0^{t_2} \prod_{k=1}^n G(t_{k+1} - t_k, x_{k+1} - x_k) dt_1 \cdots dt_n$$

where $t = t_{n+1}$ and $x = x_{n+1}$, and say that

1. $u(t, x) = 1 + \sum_{k=1}^{\infty} \theta^{k/2} I_k(f_k(\cdot, x, t)), \quad (t, x) \in (0, T) \times \mathbb{R}^d$
2. $\mathbb{E}(u(t, x)^2) = \sum_{k=0}^{\infty} \theta^k \|f_k(\cdot, x, t)\|_{\mathcal{H}^{\otimes n}}^2, \quad (t, x) \in (0, T) \times \mathbb{R}^d.$

Through a change of variable, one can easily show that

1. $\mathcal{F}G(t, \cdot)(c\xi) = c^{-\frac{a}{b}(b+r-1)} \mathcal{F}G\left(c^{\frac{a}{b}}t, \cdot\right)(\xi)$
2. $\left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 = t^{[2(b+r)-b\alpha/a]n} \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2$

How to Find the Critical α

For simplicity we consider the SWE ($a = b = 2$ and $r = 0$).

$$\begin{aligned}\|u(t, x)\|_2^2 &= \sum_{n \geq 0} \theta^n n! \overbrace{t^{[4-\alpha]n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2} \\ &= \sum_{n \geq 0} \frac{\theta^n t^{(4-\alpha)n}}{(n!)^{3-\alpha}} R_n, \quad R_n = (n!)^{4-\alpha} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2\end{aligned}$$

When $\alpha = 3$, then the above reduces down to

$$\|u(t, x)\|_2^2 = \sum_{n \geq 0} (\theta t)^n n! \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2.$$

and we lose the $n!$ term in the demonstrator.

ρ

We define

$$\rho_{\nu,a}(\gamma) = \sup_{\|f\|_{L^2(\mathbb{R}^d)}=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f(x+y)f(y)}{\sqrt{1+\frac{\nu}{2}|x+y|^a} \sqrt{1+\frac{\nu}{2}|y|^a}} dy \right]^2 \mu(dx)$$

Theorem (X. Chen '07)

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right] \\ = \log(\rho_{\nu,a}(\gamma)).$$

where γ is the spatial correlation function.

Connection of the SPDE with ρ

Recall from above that

$$\mathcal{FLG}(1, \xi) = \frac{1}{1 + \frac{\nu}{2} |\xi|^a}.$$

By using this relation, it can be shown that

$$\mathcal{FL}(\tilde{f}_n)(1, \xi) = \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a}.$$

Applying this to the limit representation of ρ above yields that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\int_{(\mathbb{R}^d)^n} \left| \mathcal{FL}(\tilde{f}_n)(1, \xi) \right|^2 \mu(d\vec{\xi}) \right] = \log(\rho_{\nu, a}(\gamma)).$$

Recall the scaling property

$$\left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 = t^{(4-\alpha)n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2.$$

Using this scaling property we see that

$$\int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt = \Gamma((4-\alpha)n+1) \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2.$$

Lemma (R. Balan, L. Chen & X. Chen '21)

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\Gamma((4-\alpha)n+1) \left\| \tilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) = \log [2^{4-\alpha} \rho]$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left((n!)^{4-\alpha} \left\| \tilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) = \log \left[\left(\frac{2}{4-\alpha} \right)^{4-\alpha} \rho \right]$$

How to Find the Blowup Time (SWE)

Recall from above that the critical case corresponds to $\alpha = 3 = d$.

$$\begin{aligned}\|u(t, x)\|_2^2 &= \sum_{n \geq 0} \theta^n n! \left\| \tilde{f}(\cdot, 0, t) \right\|_{H^{\otimes n}}^2 \\ &= \sum_{n \geq 0} (\theta t)^n n! \left\| \tilde{f}(\cdot, 0, 1) \right\|_{H^{\otimes n}}^2 \\ &=: \sum_{n \geq 0} (\theta t)^n R_n, \quad R_n = n! \left\| \tilde{f}(\cdot, 0, 1) \right\|_{H^{\otimes n}}^2.\end{aligned}$$

The above lemma says that $\frac{1}{n} \log R_n \rightarrow \log(2\rho_c)$. Now the Cauchy-Hadamard theorem can be directly applied to see that the radius of convergence is precisely $(2\theta\rho_c)^{-1} = T_2$.

We define

$$\begin{aligned}\mathcal{M}_{a,d}(\gamma, \theta) &:= \sup_{g \in \mathcal{F}_a} \left\{ \left(\iint_{\mathbb{R}^{2d}} g^2(x) g^2(y) \gamma(x+y) dx dy \right)^{1/2} - \frac{\theta}{2} \mathcal{E}_a(g, g) \right\} \\ &= \sup_{g \in \mathcal{F}_a} \left\{ \langle g^2 * g^2, \gamma \rangle_{L^2(\mathbb{R}^d)}^{1/2} - \frac{\theta}{2} \mathcal{E}_a(g, g) \right\},\end{aligned}$$

where

$$\begin{aligned}\mathcal{E}_a(g, g) &:= (2\pi)^{-d} \int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}g(\xi)|^2 d\xi \quad \text{and} \\ \mathcal{F}_a &:= \left\{ f \in L^2(\mathbb{R}^d) : \|f\|_{L^2(\mathbb{R}^d)} = 1, \mathcal{E}_a(f, f) < \infty \right\}.\end{aligned}$$

The Connection between ρ and $\mathcal{M}$

Recall that

$$\rho_{\nu,a}(\gamma) = \sup_{\|f\|_{L^2(\mathbb{R}^d)}=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f(x+y)f(y)}{\sqrt{1+\frac{\nu}{2}|x+y|^a}\sqrt{1+\frac{\nu}{2}|y|^a}} dy \right]^2 \mu(dx).$$

Theorem (Bass, X. Chen and Rosen '09)

$$\rho_{\nu,a}(\gamma) = \nu^{-\alpha/a} \mathcal{M}_a^{2-(\alpha/a)}(\gamma) < \infty$$

The Moment Asymptotic

Theorem (Chen-E. '21+)

Suppose a global solution to the SPDE in question exists and $\dot{W}$ is given through the generalized Riesz kernel defined above. Then,

$$\begin{aligned} & \lim_{t_p \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p \\ &= \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^\beta \\ & \quad \times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}\right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right), \end{aligned}$$

where

$$\beta := \frac{2(b+r) - \frac{b\alpha}{a}}{2(b+r) - \frac{b\alpha}{a} - 1} \quad \text{and} \quad t_p := (p-1)^{1-1/\beta} t.$$

Exact Moment Lyapunov Exponent

(Recall $\beta := \frac{2(b+r) - \frac{b\alpha}{a}}{2(b+r) - \frac{b\alpha}{a} - 1}$)

Corollary (Chen-E. '21+)

For $p \geq 2$ fixed, we have that

$$\begin{aligned} & \lim_{t \rightarrow \infty} t^{-\beta} \log \mathbb{E}(|u(t, x)|^p) \\ &= p(p-1)^{\frac{1}{2(b+r) - \frac{b\alpha}{a} - 1}} \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^{\beta} \\ & \quad \times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}\right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right). \end{aligned}$$

From this we can deduce that $t \mapsto \mathbb{E}(|u(t, x)|^p)$ will grow like $\exp(K_1 t^\beta)$, in other words, the moments will grow exponentially in time, and we obtain the exact expression for the constant K_1 .

Exact Large Moment Asymptotics

(Recall $\beta := \frac{2(b+r) - \frac{b\alpha}{a}}{2(b+r) - \frac{b\alpha}{a} - 1}$)

Corollary (Chen-E. '21+)

For $t > 0$ fixed, we have that

$$\begin{aligned} & \lim_{p \rightarrow \infty} p^{-\beta} \log \mathbb{E}(|u(t, x)|^p) \\ &= t^\beta \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^\beta \\ & \quad \times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}\right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right). \end{aligned}$$

From this we can deduce that $p \mapsto \mathbb{E}(|u(t, x)|^p)$ will grow like $\exp(K_2 p^\beta)$, in other words, the moments will grow exponentially in p as well.

End

Thank you for listening!

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Anticipated Question: Finiteness of ρ

Recall $\varphi(x) = \prod_{i=1}^n |x_{(i)}|^{d_i - \alpha_i}$. By examining Lemma 1.6 (X. Chen 2007), to show ρ is finite for the generalized Riesz kernel, then we need to prove the following:

$$\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} F(y) G(x) \varphi(x - y) dy dx \leq C \|F\|_{2d/(d+\alpha)} \|G\|_{2d/(d+\alpha)}$$

This can be done by using weak Young's inequality

$$\left| \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} a(x) b(x - y) c(y) dx dy \right| \leq K_{p,q,r,d} \|a\|_p \|b\|_{q,w} \|c\|_r$$

where $q^{-1} + q'^{-1} = 1$ and $p^{-1} + q^{-1} + r^{-1} = 2$ and

$$\|b\|_{q,w} = \sup_A |A|^{-1/q'} \int_A |b(x)| dx, \quad |A| < \infty.$$

We apply this with

$$a = F, \quad b = \varphi, \quad c = G$$

and we show that $\|\varphi\|_{q,w} < \infty$ with $q = d/(d - \alpha)$.

Anticipated Question: Why does ρ appear in the Lemma.

For simplicity we will consider the second moments of stochastic wave equation ($a = b = 2$ and $r = 0$). We will also only consider the limit

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\Gamma((4 - \alpha)n + 1) \left\| \tilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) = \log [2^{4-\alpha} \rho]$$

Recall that

$$\int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt = \Gamma((4 - \alpha)n + 1) \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2.$$

Define $T_n := \int_{(\mathbb{R}^d)^n} \left| \mathcal{FL} \tilde{f}_n(1, \xi) \right|^2 \mu(d\xi)$. Through direct calculation and Sterlings formula:

$$\frac{c_\alpha}{C_n} 2^{(4-\alpha)n} T_n \leq \int_0^\infty e^{-t} \left\| \tilde{f}(\cdot, 0, t) \right\|_{H^{\otimes n}}^2 dt \leq 2^{(4-\alpha)n} T_n$$

Where $\log(C_n)/n \rightarrow 0$ and $c_\alpha > 0$.

Anticipated Question: Finding the moment asymptotic

$$\mathbb{E}(|u(t, x)|^2) = \sum_{n \geq 0} z_n R_n t^{(2(b+r)-b\alpha/a)n}$$

$$R_n = (n!)^{2(b+r)-b\alpha/a} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \quad \text{and} \quad z_n = \frac{\theta^n}{(n!)^{2(b+r)-(b\alpha/a)-1}}.$$

$$\frac{1}{n} \log(R_n) \rightarrow \log \left(\left(\frac{2}{2(b+r) - \frac{b\alpha}{a}} \right)^{2(b+r) - \frac{b\alpha}{a}} \rho \right) \quad \text{as } n \rightarrow \infty.$$

If we find a β and A so that

$$\lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \sum_{n \geq 0} z_n R_n \left(t^{(2(b+r)-b\alpha/a)} \right)^n = A.$$

then this implies

$$\lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \sum_{n \geq 0} z_n R_n \left(t^{(2(b+r)-b\alpha/a)} \right)^n = A.$$